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Chapter 9

Interest

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Exercise 9(A)

Question 1(i)

The interest on ₹ 483 for 2 years at 5% per annum is:

1. ₹ 4,830

2. ₹ 48.30

3. ₹ 4.83

4. ₹ 96.60



Answer**Given:**

$$P = ₹ 483$$

$$R = 5\%$$

$$T = 2 \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{483 \times 5 \times 2}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{4830}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ 48.3$$

Hence, option 2 is the correct option.

Question 1(ii)

The simple interest on ₹ 8,490 at 5% and 73 days is

1. ₹ 849
2. ₹ 84.90
3. ₹ 8.49
4. none of these

Answer

Given:

$$P = ₹ 8,490$$

$$R = 5\%$$

$$T = 73 \text{ days} = \frac{73}{365} \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{8,490 \times 5 \times 73}{100 \times 365} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{30,98,850}{36500} \right)$$

$$\Rightarrow \text{S.I.} = ₹ 84.9$$

Hence, option 2 is the correct option.

Question 1(iii)

A sum of money, put at simple interest doubles itself in 8 years. The same sum will triple itself in:

1. 16 years
2. 12 years
3. 24 years
4. 18 years

Answer

Given:

$$A = 2P$$

$$T = 8 \text{ years}$$

Let the rate be r .

$$\therefore A = \text{S.I.} + P$$

$$\Rightarrow \text{S.I.} = A - P$$

$$\Rightarrow \text{S.I.} = 2P - P$$

$$\Rightarrow \text{S.I.} = P$$

And we know,

$$\text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow P = ₹ \left(\frac{P \times r \times 8}{100} \right)$$

$$\Rightarrow \cancel{P} = ₹ \left(\frac{\cancel{P} \times r \times 8}{100} \right)$$

$$\Rightarrow r = \frac{100}{8} \Rightarrow r = \frac{25}{2}$$

When A becomes 3P, $\Rightarrow \text{S.I.} = 2P$

Let the time be t years

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 2P = ₹ \left(\frac{P \times 25 \times t}{2 \times 100} \right)$$

$$\Rightarrow 2\cancel{P} = ₹ \left(\frac{\cancel{P} \times 25 \times t}{200} \right)$$

$$\Rightarrow t = ₹ \frac{400}{25}$$

$$\Rightarrow t = 16$$

Hence, option 1 is the correct option.

Question 1(iv)

₹ 5,000 earns ₹ 500 as simple interest in 2 years. Then the rate of interest is

1. 10%
2. 5%
3. 20%
4. 2%

Answer

Given:

$$P = ₹ 5,000$$

$$T = 2 \text{ years}$$

$$\text{S.I.} = ₹ 500$$

Let rate of interest be r .

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 500 = ₹ \left(\frac{5,000 \times r \times 2}{100} \right)$$

$$\Rightarrow 500 = ₹ \frac{10,000r}{100}$$

$$\Rightarrow 500 = ₹ 100r$$

$$\Rightarrow r = \frac{500}{100}$$

$$\Rightarrow r = 5$$

Hence, option 2 is the correct option.

Question 1(v)

₹ 7,000 earns ₹ 1,400 as interest at 5% per annum. Then the time in this case is:

1. 5 years
2. 2 years
3. 10 years
4. 4 years

Answer

Given:

$$P = ₹ 7,000$$

$$R = 5\%$$

$$S.I. = ₹ 1,400$$

Let time be t years.

$$\therefore S.I. = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 1,400 = ₹ \left(\frac{7,000 \times 5 \times t}{100} \right)$$

$$\Rightarrow 1,400 = ₹ \frac{35,000t}{100}$$

$$\Rightarrow 1,400 = ₹ 350t$$

$$\Rightarrow t = \frac{1,400}{350}$$

$$\Rightarrow t = 4$$

$\therefore$ Time = 4 years

Hence, option 4 is the correct option.

Question 2(i)

Find the interest and the amount on:

₹ 750 in 3 years 4 months at 10% per annum.

Answer**Given:**

$$P = ₹ 750$$

$$R = 10\%$$

$$T = 3 \text{ years } 4 \text{ months}$$

$$= \left(3 + \frac{4}{12}\right) \text{ years}$$

$$= \left(3 + \frac{1}{3}\right) \text{ years}$$

$$= \frac{10}{3} \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{750 \times 10 \times 10}{3 \times 100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{75,000}{300} \right)$$

$$\Rightarrow \text{S.I.} = ₹ 250$$

$$\therefore A = P + \text{S.I.}$$

$$\Rightarrow A = ₹ 750 + ₹ 250$$

$$\Rightarrow A = ₹ 1,000$$

Hence, S.I. = ₹ 250 and Amount = ₹ 1,000.

Question 2(ii)

Find the interest and the amount on:

₹ 5,000 at 8% per year from 23rd December 2011 to 29th July 2012.

Answer

Given:

$$P = ₹ 5,000$$

$$R = 8\%$$

To calculate time (T):

$$\text{Dec} = 8 \text{ days } (31 - 23)$$

$$\text{Jan} = 31 \text{ days}$$

$$\text{Feb} = 29 \text{ days}$$

$$\text{March} = 31 \text{ days}$$

$$\text{April} = 30 \text{ days}$$

$$\text{May} = 31 \text{ days}$$

$$\text{June} = 30 \text{ days}$$

$$\text{July} = 29 \text{ days}$$

$$\text{Total} = 219 \text{ days}$$

$$T = 219 \text{ days}$$

$$= \left(\frac{219}{365} \right) \text{ years}$$

$$= \left(\frac{3}{5} \right) \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{5,000 \times 8 \times 3}{5 \times 100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{1,20,000}{500}$$

$$\Rightarrow \text{S.I.} = ₹ 240$$

$$\therefore A = P + \text{S.I.}$$

$$\Rightarrow A = ₹ 5,000 + ₹ 240$$

$$\Rightarrow A = ₹ 5,240$$

Hence, S.I. = ₹ 240 and Amount = ₹ 5,240.

Question 2(iii)

Find the interest and the amount on:

₹ 2,600 in 2 years 3 months at 1% per month.

Answer

Given:

$$P = ₹ 2,600$$

$$R = 1\% \text{ per month} = 12\% \text{ per annum}$$

$$T = 2 \text{ years } 3 \text{ months}$$

$$= \left(2 + \frac{3}{12}\right) \text{ years}$$

$$= \left(2 + \frac{1}{4}\right) \text{ years}$$

$$= \left(\frac{8+1}{4}\right) \text{ years}$$

$$= \left(\frac{9}{4}\right) \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{2,600 \times 12 \times 9}{4 \times 100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{2,08,800}{400}$$

$$\Rightarrow \text{S.I.} = ₹ 702$$

$$\therefore A = P + \text{S.I.}$$

$$\Rightarrow A = ₹ 2,600 + ₹ 702$$

$$\Rightarrow A = ₹ 3,302$$

Hence, S.I. = ₹ 702 and Amount = ₹ 3,302.

Question 2(iv)

Find the interest and the amount on:

₹ 4,000 in $1\frac{1}{3}$ years at 2 paise per rupee per month.

Answer

Given:

$$P = ₹ 4,000$$

$$R = 2 \text{ paise per rupee per month}$$

$$= \frac{2}{100} \text{ per month}$$

$$= \frac{2}{100} \times 12 \text{ per annum}$$

$$= \frac{24}{100} \text{ per annum}$$

$$= 24\% \text{ per annum}$$

$$T = \left(1\frac{1}{3}\right) \text{ years}$$

$$= \left(\frac{4}{3}\right) \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{4,000 \times 24 \times 4}{3 \times 100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{3,84,000}{300}$$

$$\Rightarrow \text{S.I.} = ₹ 1,280$$

$$\therefore A = P + \text{S.I.}$$

$$\Rightarrow A = ₹ 4,000 + ₹ 1,280$$

$$\Rightarrow A = ₹ 5,280$$

Hence, S.I. = ₹ 1,280 and Amount = ₹ 5,280.

Question 3

Rohit borrowed ₹ 24,000 at 7.5 percent per year. How much money will he pay at the end of 4 years to clear his debt?

Answer**Given:**

$$P = ₹ 24,000$$

$$R = 7.5\%$$

$$T = 4 \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{24,000 \times 7.5 \times 4}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{7,20,000}{100}$$

$$\Rightarrow \text{S.I.} = ₹ 7,200 \text{ And } \therefore A = P + \text{S.I.}$$

$$\Rightarrow A = ₹ 24,000 + ₹ 7,200$$

$$\Rightarrow A = ₹ 31,200$$

Hence, Rohit will pay ₹ 31,200 at the end of 4 years.

Question 4

On what principal will the simple interest be ₹ 7,008 in 6 years 3 months at 5% per year?

Answer

Given:

$$\text{S.I.} = ₹ 7,008$$

$$R = 5\%$$

$$T = 6 \text{ years } 3 \text{ months}$$

$$= \left(6 + \frac{3}{12}\right) \text{ years}$$

$$= \left(6 + \frac{1}{4}\right) \text{ years}$$

$$= \frac{25}{4} \text{ years}$$

Let the Principal amount be ₹ P .

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 7,008 = \left(\frac{P \times 5 \times 25}{4 \times 100} \right)$$

$$\Rightarrow 7,008 = \left(\frac{P \times 125}{400} \right)$$

$$\Rightarrow P = \left(\frac{7,008 \times 400}{125} \right)$$

$$\Rightarrow P = \frac{28,03,200}{125}$$

$$\Rightarrow P = 22,425.60$$

Hence, the Principal amount be ₹ 22,425.60.

Question 5

Find the principal which will amount to ₹ 4,000 in 4 years at 6.25% per annum.

Answer

Given:

$$A = ₹ 4,000$$

$$R = 6.25\%$$

$$T = 4 \text{ years}$$

Let the Principal amount be ₹ P .

$$\text{As we know, } A = \text{S.I.} + P$$

$$\Rightarrow 4,000 = \text{S.I.} + P$$

$$\Rightarrow \text{S.I.} = 4,000 - P$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 4,000 - P = \left(\frac{P \times 6.25 \times 4}{100} \right)$$

$$\Rightarrow 4,000 - P = \frac{P \times 25}{100}$$

$$\Rightarrow 4,000 - P = \frac{P}{4}$$

$$\Rightarrow 4,000 = \frac{P}{4} + P$$

$$\Rightarrow 4,000 = \frac{P}{4} + \frac{4P}{4}$$

$$\Rightarrow 4,000 = \frac{(P + 4P)}{4}$$

$$\Rightarrow 4,000 = \frac{5P}{4}$$

$$\Rightarrow P = \frac{4,000 \times 4}{5}$$

$$\Rightarrow P = \frac{16,000}{5}$$

$$\Rightarrow P = 3,200$$

Hence, the Principal amount be ₹ 3,200.

Question 6(i)

At what rate per cent per annum will ₹ 630 produce an interest of ₹ 126 in 4 years?

Answer

Given:

$$P = ₹ 630$$

$$T = 4 \text{ years}$$

$$\text{S.I.} = ₹ 126$$

Let rate of interest be r .

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 126 = ₹ \left(\frac{630 \times r \times 4}{100} \right)$$

$$\Rightarrow 126 = ₹ \frac{2,520r}{100}$$

$$\Rightarrow 126 = ₹ \frac{252r}{10}$$

$$\Rightarrow r = \frac{126 \times 10}{252}$$

$$\Rightarrow r = 5$$

Hence, the rate of interest = 5%.

Question 6(ii)

At what rate percent per year will a sum double itself in $6\frac{1}{4}$ years?

Answer

Given:

$$T = 6\frac{1}{4} \text{ years}$$

$$= \frac{25}{4} \text{ years}$$

$$A = 2P$$

$$A = P + \text{S.I.}$$

$$\Rightarrow 2P = P + \text{S.I.}$$

$$\Rightarrow 2P - P = \text{S.I.}$$

$$\Rightarrow P = \text{S.I.}$$

Let rate of interest be r .

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow P = ₹ \left(\frac{P \times r \times 25}{4 \times 100} \right)$$

$$\Rightarrow \cancel{P} = ₹ \frac{\cancel{P} \times r \times 25}{400}$$

$$\Rightarrow 1 = ₹ \frac{25r}{400}$$

$$\Rightarrow 400 = 25r$$

$$\Rightarrow r = \frac{400}{25}$$

$$\Rightarrow r = 16$$

Hence, the rate of interest = 16%.

Question 6(iii)

Find the rates of interest per year, if the interest charged for 8 months be 0.06 times of the money borrowed.

Answer

Given:

T = 8 months

$$= \frac{8}{12} \text{ years}$$

$$= \frac{2}{3} \text{ years}$$

Let principal be P

S.I. = 0.06 P

Let rate of interest be r .

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 0.06P = ₹ \left(\frac{P \times r \times 2}{3 \times 100} \right)$$

$$\Rightarrow \frac{6P}{100} = ₹ \left(\frac{P \times r \times 2}{300} \right)$$

$$\Rightarrow \frac{6}{100} \cancel{P} = ₹ \left(\frac{\cancel{P} \times r \times 2}{300} \right)$$

$$\Rightarrow \frac{6}{100} = \frac{2r}{300}$$

$$\Rightarrow r = \frac{6 \times 300}{100 \times 2}$$

$$\Rightarrow r = \frac{1800}{200}$$

$$\Rightarrow r = 9$$

Hence, the rate of interest = 9%.

Question 7(i)

In how many years will ₹ 950 produce ₹ 399 as simple interest at 7% ?

Answer

Given:

$$P = ₹ 950$$

$$R = 7\%$$

$$S.I. = ₹ 399$$

$$\text{Let time be } t \text{ years. } \therefore S.I. = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 399 = ₹ \left(\frac{950 \times 7 \times t}{100} \right)$$

$$\Rightarrow 399 = ₹ \left(\frac{6,650t}{100} \right)$$

$$\Rightarrow 399 = ₹\left(\frac{133t}{2}\right)$$

$$\Rightarrow t = \frac{399 \times 2}{133}$$

$$\Rightarrow t = \frac{798}{133}$$

$$\Rightarrow t = 6$$

Hence, the time is 6 years.

Question 7(ii)

Find the time in which ₹ 1,200 will amount to ₹ 1,536 at 3.5% per year.

Answer

Given:

$$P = ₹ 1,200$$

$$R = 3.5\%$$

$$A = ₹ 1,536$$

As we know,

$$\therefore A = P + \text{S.I.}$$

$$\Rightarrow ₹ 1,536 = ₹ 1,200 + \text{S.I.}$$

$$\Rightarrow \text{S.I.} = ₹ 1,536 - ₹ 1,200$$

$$\Rightarrow \text{S.I.} = ₹ 336$$

Let time be t years.

$$\therefore \text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 336 = \left(\frac{1,200 \times 3.5 \times t}{100} \right)$$

$$\Rightarrow 336 = \left(\frac{4,200t}{100} \right)$$

$$\Rightarrow 336 = 42t$$

$$\Rightarrow t = \frac{336}{42}$$

$$\Rightarrow t = 8$$

Hence, the time is 8 years.

Question 8

The simple interest on a certain sum of money is $\frac{3}{8}$ of the sum in $6\frac{1}{4}$ years.

Find the rate percent charged.

Answer

Given:

$$T = 6\frac{1}{4} \text{ years}$$

$$= \frac{25}{4} \text{ years}$$

Let principal be P

$$\text{S.I.} = \frac{3}{8} P$$

Let rate of interest be r .

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \frac{3}{8} P = ₹ \left(\frac{P \times r \times 25}{4 \times 100} \right)$$

$$\Rightarrow \frac{3P}{8} = ₹ \left(\frac{P \times r \times 25}{400} \right)$$

$$\Rightarrow \frac{3}{8} \cancel{P} = ₹ \frac{\cancel{P} \times 25r}{400}$$

$$\Rightarrow \frac{3}{8} = ₹ \frac{25r}{400}$$

$$\Rightarrow r = \frac{3 \times 400}{8 \times 25}$$

$$\Rightarrow r = \frac{1200}{200}$$

$$\Rightarrow r = 6$$

Hence, the rate of interest = 6%.

Question 9

What sum of money borrowed on 24th May will amount to ₹ 10,210.20 on 17th October of the same year at 5 percent per annum simple interest?

Answer

$$A = ₹ 10,210.20$$

$$R = 5\%$$

To calculate time (T):

$$\text{May} = 7 \text{ days } (31 - 24)$$

$$\text{Jun} = 30 \text{ days}$$

$$\text{July} = 31 \text{ days}$$

$$\text{August} = 31 \text{ days}$$

$$\text{Sept} = 30 \text{ days}$$

$$\text{Oct} = 17 \text{ days}$$

$$\text{Total} = 146 \text{ days}$$

$$T = 146 \text{ days}$$

$$= \frac{146}{365} \text{ years}$$

$$= \frac{2}{5} \text{ years}$$

Let the Principal amount be ₹ P .

As we know,

$$A = \text{S.I.} + P$$

$$\Rightarrow 10,210.20 = \text{S.I.} + P$$

$$\Rightarrow \text{S.I.} = 10,210.20 - P$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 10,210.20 - P = \left(\frac{P \times 5 \times 2}{5 \times 100} \right)$$

$$\Rightarrow 10,210.20 - P = \left(\frac{P \times 2}{100} \right)$$

$$\Rightarrow 10,210.20 - P = \frac{P}{50}$$

$$\Rightarrow 10,210.20 = \frac{P}{50} + P$$

$$\Rightarrow 10,210.20 = \frac{P}{50} + \frac{50P}{50}$$

$$\Rightarrow 10,210.20 = \frac{(P + 50P)}{50}$$

$$\Rightarrow 10,210.20 = \frac{51P}{50}$$

$$\Rightarrow P = \frac{10,210.20 \times 50}{51}$$

$$\Rightarrow P = \frac{510510}{51}$$

$$\Rightarrow P = 10,010$$

Hence, the Principal amount be ₹ 10,010.

Question 10

In what time will the interest on a certain sum of money at 6% be $\frac{5}{8}$ of itself?

Answer

Let the Principal amount be ₹ P.

$$\text{S.I.} = \frac{5}{8} \text{ of } P$$

$$R = 6\%$$

Let time be t years.

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \frac{5}{8} \times P = ₹ \left(\frac{P \times 6 \times t}{100} \right)$$

$$\Rightarrow \frac{5}{8} \times \cancel{P} = ₹ \left(\frac{\cancel{P} \times 3t}{50} \right)$$

$$\Rightarrow \frac{5}{8} = ₹ \left(\frac{3t}{50} \right)$$

$$\Rightarrow t = \frac{5 \times 50}{8 \times 3}$$

$$\Rightarrow t = \frac{250}{24}$$

$$\Rightarrow t = 10\frac{10}{24}$$

$$\Rightarrow t = 10\frac{5}{12}$$

Hence, the time is 10 years and 5 months.

Question 11

Ashok lent out ₹ 7,000 at 6% and ₹ 9,500 at 5%. Find his total income from the interest in 3 years.

Answer

Given:

$$P = ₹ 7,000$$

$$R = 6\%$$

$$T = 3 \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{7,000 \times 6 \times 3}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{1,26,000}{100}$$

$$\Rightarrow \text{S.I.} = ₹ 1,260$$

$$P = ₹ 9,500$$

$$R = 5\%$$

$$T = 3 \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{9,500 \times 5 \times 3}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{1,42,500}{100}$$

$$\Rightarrow \text{S.I.} = ₹ 1,425$$

Total income from the interest = ₹ 1,260 + ₹ 1,425 = ₹ 2,685

Hence, total amount from the interest in 3 years = ₹ 2,685.

Question 12

Raj borrows ₹ 8,000; out of which ₹ 4,500 at 5% and the remaining at 6%. Find the total interest paid by him in 4 years.

Answer

Given:

Total amount Raj borrows = ₹ 8,000

P = ₹ 4,500

R = 5%

T = 4 years

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{4,500 \times 5 \times 4}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{90,000}{100}$$

$$\Rightarrow \text{S.I.} = ₹900$$

$$\text{Remaining Principal} = ₹ (8,000 - 4500) = ₹ 3,500$$

$$R = 6\%$$

$$T = 4 \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{3,500 \times 6 \times 4}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{84,000}{100}$$

$$\Rightarrow \text{S.I.} = ₹840$$

$$\text{Total interest paid in 4 years} = ₹(900 + 840) = ₹ 1,740$$

Hence, total interest paid in 4 years = ₹ 1,740.

Exercise 9(B)

Question 1(i)

₹ 5,000 is put in a bank at 5% simple interest. The amount at the end of 2 years will be :

1. ₹ 250
2. ₹ 500
3. ₹ 5,500
4. ₹ 4,500

Answer

Given:

$$P = ₹ 5,000$$

$$R = 5\%$$

$$T = 2 \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{5,000 \times 5 \times 2}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{50,000}{100}$$

$$\Rightarrow \text{S.I.} = ₹500$$

And,

$$A = P + \text{S.I.}$$

$$\Rightarrow A = ₹5,000 + ₹500$$

$$\Rightarrow A = ₹5,500$$

Hence, option 3 is the correct option.

Question 1(ii)

A sum of money triples itself in 20 years. The rate of interest is:

1. 20%
2. 10%
3. 5%
4. 15%

Answer

Given:

$$A = 3P$$

$$T = 20 \text{ years}$$

Let the rate be r .

$$\therefore A = \text{S.I.} + P$$

$$\Rightarrow \text{S.I.} = A - P$$

$$\Rightarrow \text{S.I.} = 3P - P$$

$$\Rightarrow \text{S.I.} = 2P$$

And we know,

$$\text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 2P = ₹ \left(\frac{P \times r \times 20}{100} \right)$$

$$\Rightarrow 2\cancel{P} = ₹ \frac{\cancel{P} \times r \times 20}{100}$$

$$\Rightarrow 2 = ₹ \frac{r}{5}$$

$$\Rightarrow r = 2 \times 5$$

$$\Rightarrow r = 10$$

Hence, option 2 is the correct option.

Question 1(iii)

A sum of money earns simple interest equal to 0.5 times the sum in 10 years;
the rate of interest per annum is :

1. 20%
2. 10%
3. 5%
4. none of these

Answer

Given:

$$\text{S.I.} = 0.5P$$

$$= \frac{5}{10} P$$

$$T = 10 \text{ years}$$

Let the rate be r .

We know,

$$\text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \frac{5}{10} P = ₹ \left(\frac{P \times r \times 10}{100} \right)$$

$$\Rightarrow \frac{1}{2} \cancel{P} = ₹ \frac{\cancel{P} \times R \times 10}{100}$$

$$\Rightarrow \frac{1}{2} = ₹ \frac{R}{10}$$

$$\Rightarrow R = \frac{10}{2}$$

$$\Rightarrow R = 5$$

Hence, option 3 is the correct option.

Question 1(iv)

A sum of ₹ 600 put at 5% S.I. amounts to ₹ 720 in:

1. 3 years
2. 4 years
3. 5 years
4. 6 years

Answer

Given:

$$P = ₹ 600$$

$$R = 5\%$$

$$A = ₹ 720$$

Let the time be t .

$$A = P + \text{S.I.}$$

$$\Rightarrow ₹720 = ₹600 + \text{S.I.}$$

$$\Rightarrow \text{S.I.} = ₹720 - ₹600$$

$$\Rightarrow \text{S.I.} = ₹120$$

And

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 120 = \left(\frac{600 \times 5 \times t}{100} \right)$$

$$\Rightarrow 120 = \frac{3,000 \times t}{100}$$

$$\Rightarrow 120 = 30 \times t$$

$$\Rightarrow t = \frac{120}{30}$$

$$\Rightarrow t = 4$$

Hence, option 2 is the correct option.

Question 1(v)

Manoj invested ₹ 8,000 for 10 years at 10% p.a. simple interest. The amount at the end of 2 years will be :

1. ₹ 9,600
2. ₹ 9,800
3. ₹ 16,000
4. None of these

Answer

Given:

$$P = ₹ 8,000$$

$$R = 10\%$$

$$T = 2 \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{8,000 \times 10 \times 2}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{1,60,000}{100}$$

$$\Rightarrow \text{S.I.} = ₹ 1,600$$

And,

$$A = P + \text{S.I.}$$

$$\Rightarrow A = ₹ 8,000 + ₹ 1,600$$

$$\Rightarrow A = ₹ 9,600$$

Hence, option 1 is the correct option.

Question 2

If ₹ 3,750 amounts to ₹ 4,620 in 3 years at simple interest. Find:

(i) the rate of interest.

(ii) the amount of ₹ 7,500 in $5\frac{1}{2}$ years at the same rate of interest.

Answer

(i) **Given:**

$$P = ₹ 3,750$$

$$A = ₹ 4,620$$

$$T = 3 \text{ years}$$

Let the rate of interest be r .

$$A = P + \text{S.I.}$$

$$\Rightarrow ₹4,620 = ₹3,750 + \text{S.I.}$$

$$\Rightarrow \text{S.I.} = ₹4,620 - ₹3,750$$

$$\Rightarrow \text{S.I.} = ₹870$$

And,

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow ₹870 = \left(\frac{₹3,750 \times r \times 3}{100} \right)$$

$$\Rightarrow ₹870 = \frac{11,250r}{100}$$

$$\Rightarrow ₹870 = \frac{225r}{2}$$

$$\Rightarrow r = \frac{2 \times 870}{225}$$

$$\Rightarrow r = \frac{1740}{225}$$

$$\Rightarrow r = \frac{116}{15}$$

$$\Rightarrow r = 7\frac{11}{15}$$

Hence, the rate of interest = $7\frac{11}{15}$.

(ii) Given:

$$P = ₹ 7,500$$

$$R = \frac{116}{15}$$

$$T = 5\frac{1}{2} \text{ years}$$

$$= \frac{11}{2} \text{ years}$$

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \left(\frac{7,500 \times 116 \times 11}{15 \times 2 \times 100} \right)$$

$$\Rightarrow \text{S.I.} = ₹ \frac{95,70,000}{3000}$$

$$\Rightarrow \text{S.I.} = ₹ 3,190$$

And,

$$A = P + \text{S.I.}$$

$$\Rightarrow A = ₹7,500 + ₹3,190$$

$$\Rightarrow A = ₹10,690$$

Hence, the amount = ₹ 10,690.

Question 3

A sum of money, lent out at simple interest, doubles itself in 8 years. Find:

(i) the rate of interest.

(ii) in how many years will the sum become triple (three times) of itself at the same rate percent?

Answer

(i) Let the Principal amount be ₹ P.

$$A = 2P$$

$$T = 8 \text{ years}$$

Let the rate be r .

$$\therefore A = \text{S.I.} + P$$

$$\Rightarrow \text{S.I.} = A - P$$

$$\Rightarrow \text{S.I.} = 2P - P$$

$$\Rightarrow \text{S.I.} = P$$

And we know,

$$\text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow P = ₹ \left(\frac{P \times r \times 8}{100} \right)$$

$$\Rightarrow \cancel{P} = ₹ \frac{\cancel{P} \times r \times 8}{100}$$

$$\Rightarrow 1 = ₹ \frac{8r}{100}$$

$$\Rightarrow r = \frac{100}{8} \Rightarrow r = \frac{25}{2} \Rightarrow r = 12.5$$

Hence, rate of interest = 12.5.

(ii) When A becomes 3P, $\Rightarrow \text{S.I.} = 2P$

Let the time be t years

$$\therefore \text{S.I.} = ₹ \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 2P = ₹ \left(\frac{P \times 25 \times t}{2 \times 100} \right)$$

$$\Rightarrow 2\cancel{P} = ₹ \frac{\cancel{P} \times 25 \times t}{200}$$

$$\Rightarrow t = \frac{400}{25}$$

$$\Rightarrow t = 16$$

Hence, the time is 16 years.

Question 4

Rupees 4,000 amounts to ₹ 5,000 in 8 years; in what time will ₹ 2,100 amount to ₹ 2,800 at the same rate?

Answer

Given:

$$P = ₹ 4,000$$

$$A = ₹ 5,000$$

$$T = 8 \text{ years}$$

Let the rate be r .

$$\text{As we know, } A = P + S.I.$$

$$\Rightarrow ₹5,000 = ₹4,000 + S.I.$$

$$\Rightarrow S.I. = ₹5,000 - ₹4,000$$

$$\Rightarrow S.I. = ₹1,000$$

And

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow ₹1,000 = \left(\frac{₹4,000 \times r \times 8}{100} \right)$$

$$\Rightarrow ₹1,000 = 40 \times r \times 8$$

$$\Rightarrow ₹1,000 = 320 \times r$$

$$\Rightarrow r = \frac{1,000}{320} \Rightarrow r = \frac{25}{8}$$

When $P = ₹2,100$

$$A = ₹2,800$$

$$R = \frac{25}{8}$$

Let the time be t .

As we know,

$$A = P + \text{S.I.}$$

$$\Rightarrow ₹2,800 = ₹2,100 + \text{S.I.}$$

$$\Rightarrow \text{S.I.} = ₹2,800 - ₹2,100$$

$$\Rightarrow \text{S.I.} = ₹700$$

And

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 700 = \left(\frac{2,100 \times 25 \times t}{8 \times 100} \right)$$

$$\Rightarrow 700 = \frac{21 \times 25 \times t}{8}$$

$$\Rightarrow 700 = \frac{525 \times t}{8}$$

$$\Rightarrow t = \frac{700 \times 8}{525}$$

$$\Rightarrow t = \frac{32}{3}$$

$$\Rightarrow t = 10\frac{2}{3}$$

$$\Rightarrow t = 10\frac{8}{12}$$

Hence, the time = 10 years and 8 months.

Question 5

What sum of money lent at 6.5% per annum will produce the same interest in 4 years as ₹ 7,500 produce in 6 years at 5% per annum?

Answer

Given:

$$P_1 = ₹ 7,500$$

$$R_1 = 5\%$$

$$T_1 = 6 \text{ years}$$

$$\text{So, S.I.}_1 =$$

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$= \left(\frac{7,500 \times 5 \times 6}{100} \right)$$

$$= \frac{2,25,000}{100}$$

$$= 2,250$$

$$R_2 = 6.5\%$$

$$T_2 = 4 \text{ years}$$

Let P_2 be P .

$$\text{And S.I.}_2 =$$

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$= \left(\frac{P \times 6.5 \times 4}{100} \right)$$

$$= \frac{26P}{100}$$

As interest on both are same. Hence,

$$S.I._1 = S.I._2$$

$$\therefore 2,250 = \frac{26P}{100}$$

$$\Rightarrow P = \frac{2,250 \times 100}{26}$$

$$\Rightarrow P = \frac{2,25,000}{26}$$

$$\Rightarrow P = 8,653.85$$

Hence, the principal amount = ₹ 8,653.85.

Question 6

A certain sum amounts to ₹ 3,825 in 4 years and to ₹ 4,050 in 6 years. Find the rate percent and the sum.

Answer

Given:

Amount in 4 years = ₹ 3,825

$$\Rightarrow P + S.I. \text{ of 4 years} = ₹ 3,825 \dots\dots\dots(1)$$

Amount in 6 years = ₹ 4,050

$$\Rightarrow P + S.I. \text{ of 6 years} = ₹ 4,050 \dots\dots\dots(2)$$

Subtracting equation (1) from (2), we get

$$S.I. \text{ of 2 years} = ₹4,050 - ₹3,825$$

$$= ₹225$$

$$\text{S.I. of 1 year} = ₹ \frac{225}{2}$$

$$= ₹112.5$$

$$\text{S.I. of 4 years} = ₹112.5 \times 4$$

$$= ₹450$$

From equation (1), we get:

$$P + ₹ 450 = ₹ 3,825$$

$$P = ₹ 3,825 - ₹ 450$$

$$P = ₹ 3,375$$

$$\text{Now when } P = ₹ 3,375$$

$$\text{S.I.} = ₹ 112.5$$

$$T = 1 \text{ year}$$

Let the rate be r .

As we know

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 112.5 = \left(\frac{3,375 \times r \times 1}{100} \right)$$

$$\Rightarrow 112.5 = \frac{3,375r}{100}$$

$$\Rightarrow r = \frac{112.5 \times 100}{3375}$$

$$\Rightarrow r = \frac{11250}{3375}$$

$$\Rightarrow r = \frac{10}{3}$$

$$\Rightarrow r = 3\frac{1}{3}$$

Hence, principal = ₹3,375 and rate = $3\frac{1}{3}$

Question 7

At what rate per cent of simple interest will the interest on ₹ 3,750 be one-fifth of itself in 4 years? What will it amount to in 15 years?

Answer

Given:

$$P = ₹ 3,750$$

$$T = 4 \text{ years}$$

$$\text{S.I.} = \text{one-fifth of } P$$

$$= \frac{1}{5} \times ₹3,750$$

$$= \frac{₹3,750}{5}$$

$$= ₹750$$

Let the rate be r .

As we know,

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 750 = \left(\frac{3,750 \times r \times 4}{100} \right)$$

$$\Rightarrow 750 = 75 \times r \times 2$$

$$\Rightarrow 750 = 150r$$

$$\Rightarrow r = \frac{750}{150}$$

$$\Rightarrow r = 5$$

When $P = ₹ 3,750$

$R = 5\%$

$T = 15$ years

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$= \left(\frac{3,750 \times 5 \times 15}{100} \right)$$

$$= \left(\frac{2,81,250}{100} \right)$$

$$= 2,812.50$$

And,

$$A = P + S.I.$$

$$= 3,750 + 2,812.5$$

$$= 6,562.50$$

Hence, rate = 5 and the amount = ₹ 6,562.50.

Question 8

On what date will ₹ 1,950 lent on 5th January, 2011 amount to ₹ 2,125.50 at 5 percent per annum simple interest?

Answer

Given:

$$P = ₹ 1,950$$

$$A = ₹ 2,125.50$$

$$R = 5\%$$

Let the time be t

As we know,

$$A = P + S.I.$$

$$\Rightarrow ₹ 2,125.50 = ₹ 1,950 + S.I.$$

$$\Rightarrow S.I. = ₹2,125.50 - ₹1,950$$

$$\Rightarrow S.I. = ₹175.50$$

And,

$$S.I. = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 175.50 = \left(\frac{1,950 \times 5 \times t}{100} \right)$$

$$\Rightarrow 175.50 = \frac{9,750t}{100}$$

$$\Rightarrow t = \frac{175.50 \times 100}{9750}$$

$$\Rightarrow t = \frac{9}{5}$$

$$\Rightarrow t = 1\frac{4}{5}$$

$$1\frac{4}{5} \text{ years means 1 years and 292 days } (\because \frac{4}{5} \times 365 = 292).$$

As the money was lent on 5th January, 2011,

1 year from 5th January, 2011 = 5th January, 2012

And 292 days = Jan (26 days) + Feb (29 days) + March (31 days) + April (30 days) + May (31 days) + June (30 days) + July (31 days) + Aug (31 days) + Sept (30 days) + Oct (23 days)

Hence, the required date is 23rd October, 2012.

Question 9

If the interest on ₹ 2,400 is more than the interest on ₹ 2,000 by ₹ 60 in 3 years at the same rate per cent, find the rate.

Answer**Given:**

$$P_1 = ₹ 2,400$$

$$T_1 = 3 \text{ years}$$

Let the rate be r .

Hence,

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I} = \left(\frac{2,400 \times r \times 3}{100} \right)$$

$$\Rightarrow \text{S.I} = \left(\frac{7,200r}{100} \right)$$

$$\Rightarrow \text{S.I} = 72r$$

$$P_2 = ₹ 2,000$$

$$T_2 = 3 \text{ years}$$

$$R = r\%$$

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = \left(\frac{2,000 \times r \times 3}{100} \right)$$

$$\Rightarrow \text{S.I.} = \left(\frac{6,000r}{100} \right)$$

$$\Rightarrow \text{S.I.} = 60r$$

According to the question,

Difference between the two S.I. = ₹ 60

$$72r - 60r = ₹60$$

$$\Rightarrow 12r = ₹60$$

$$\Rightarrow r = \frac{60}{12}$$

$$\Rightarrow r = 5$$

Hence, the rate is 5%.

Question 10

Divide ₹ 15,600 into two parts such that the interest on one at 5 percent for 5 years may be equal to that on the other at $4\frac{1}{2}$ per cent for 6 years.

Answer

Let the first part be ₹ x and the second part be ₹ $(15,600 - x)$.

Hence

$$P_1 = ₹ x$$

$$R_1 = 5\%$$

$$T_1 = 5 \text{ years}$$

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = \left(\frac{x \times 5 \times 5}{100} \right)$$

$$\Rightarrow \text{S.I.} = \frac{25x}{100}$$

$$\Rightarrow \text{S.I.} = \frac{x}{4}$$

$$P_2 = ₹ 15,600 - x$$

$$R_2 = 4\frac{1}{2}$$

$$= \frac{9}{2}$$

$$T_2 = 6 \text{ years}$$

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = \left[\frac{(15,600 - x) \times 9 \times 6}{2 \times 100} \right]$$

$$\Rightarrow \text{S.I.} = \frac{54(15,600 - x)}{200}$$

$$\Rightarrow \text{S.I.} = \frac{27(15,600 - x)}{100}$$

Both interest are equal.

$$\frac{x}{4} = \frac{27(15,600 - x)}{100}$$

$$\Rightarrow \frac{x}{4} = \frac{4,21,200 - 27x}{100}$$

$$\Rightarrow \frac{x}{4} + \frac{27x}{100} = \frac{4,21,200}{100}$$

$$\Rightarrow \frac{25x}{100} + \frac{27x}{100} = 4,212$$

$$\Rightarrow \frac{(25x + 27x)}{100} = 4,212$$

$$\Rightarrow \frac{52x}{100} = 4,212$$

$$\Rightarrow x = \frac{4,212 \times 100}{52}$$

$$\Rightarrow x = \frac{4,21,200}{52}$$

$$\Rightarrow x = 8,100$$

Other amount will be ₹ (15,600 - 8,100) = ₹ 7,500.

Hence, ₹ 8,100 and ₹ 7,500 are the two parts.

Question 11

Simple interest on a certain sum is $\frac{16}{25}$ of the sum. Find the rate of interest and time, if both are numerically equal.

Answer

Given:

$$\text{S.I.} = \frac{16}{25} \text{ of } P$$

$$\text{Rate of interest} = \text{Time} = x$$

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \frac{16}{25}P = \left(\frac{P \times x \times x}{100} \right)$$

$$\Rightarrow \frac{16}{25}\cancel{P} = \left(\frac{\cancel{P} \times x \times x}{100} \right)$$

$$\Rightarrow x^2 = \frac{16 \times 100}{25}$$

$$\Rightarrow x^2 = 64$$

$$\Rightarrow x = \sqrt{64}$$

$$\Rightarrow x = 8$$

Hence, rate = 8% and time = 8 years.

Question 12

Divide ₹ 9,000 into two parts in such a way that S.I. on one part at 16% p.a. and in 2 years is equal to the S.I. on the other part at 6% p.a. and in 3 years.

Answer

Let the first part be ₹ x and the second part be ₹ $(9,000 - x)$.

Hence,

$$P_1 = ₹ x$$

$$R_1 = 16\%$$

$$T_1 = 2 \text{ years}$$

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = \left(\frac{x \times 16 \times 2}{100} \right)$$

$$\Rightarrow \text{S.I.} = \frac{32x}{100}$$

$$\Rightarrow \text{S.I.} = \frac{8x}{25}$$

$$P_2 = ₹ 9,000 - x$$

$$R_2 = 6$$

$$T_2 = 3 \text{ years}$$

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \text{S.I.} = \left[\frac{(9,000 - x) \times 6 \times 3}{100} \right]$$

$$\Rightarrow \text{S.I.} = \frac{18(9,000 - x)}{100}$$

$$\Rightarrow \text{S.I.} = \frac{9(9,000 - x)}{50}$$

Both interest are equal.

$$\frac{8x}{25} = \frac{9(9,000 - x)}{50}$$

$$\Rightarrow \frac{8x}{25} = \frac{81,000 - 9x}{50}$$

$$\Rightarrow \frac{8x}{25} + \frac{9x}{50} = \frac{81,000}{50}$$

$$\Rightarrow \frac{16x}{50} + \frac{9x}{50} = 1,620$$

$$\Rightarrow \frac{(16x + 9x)}{50} = 1,620$$

$$\Rightarrow \frac{25x}{50} = 1,620$$

$$\Rightarrow x = \frac{1,620 \times 50}{25}$$

$$\Rightarrow x = \frac{81,000}{25}$$

$$\Rightarrow x = 3,240$$

Other amount will be ₹ (9,000 - 3,240) = ₹ 5,760.

Hence, ₹ 3,240 and ₹ 5,760 are the two parts.

Exercise 9(C)

Question 1(i)

The C.I. on ₹ 1,000 at 20% per annum and in 2 years is:

1. ₹ 1,440
2. ₹ 1,240
3. ₹ 440
4. ₹ 220

Answer

For 1st year:

$$P = ₹ 1,000$$

$$R = 20\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{1,000 \times 20 \times 1}{100} \right)$$

$$= ₹ \frac{20,000}{100}$$

$$= ₹ 200$$

And

$$\text{Amount} = P + \text{Interest}$$

$$= ₹ 1,000 + 200$$

$$= ₹ 1,200$$

For 2nd year:

$$P = ₹ 1,200$$

$$R = 20\%$$

$$T = 1 \text{ year}$$

$$\begin{aligned}\text{Interest} &= \left(\frac{P \times R \times T}{100} \right) \\ &= ₹ \left(\frac{1,200 \times 20 \times 1}{100} \right) \\ &= ₹ \frac{24,000}{100} \\ &= ₹240\end{aligned}$$

And

$$\text{Final amount} = P + \text{Interest}$$

$$= ₹1,200 + 240$$

$$= ₹1,440$$

And

$$\text{Compound Interest} = \text{Final amount} - \text{Original Principal}$$

$$= ₹1,440 - ₹1,000$$

$$= ₹440$$

Hence, option 3 is the correct option.

Question 1(ii)

A sum of ₹ 2,000 is put at 10% compound interest. The amount at the end of 2 years will be :

$$1. ₹ 400$$

2. ₹ 420

3. ₹ 2420

4. ₹ 4,840

Answer

For 1st year:

$$P = ₹ 2,000$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{2,000 \times 10 \times 1}{100} \right)$$

$$= ₹ \frac{20,000}{100}$$

$$= ₹ 200$$

And

$$\text{Amount} = P + \text{Interest}$$

$$= ₹ 2,000 + 200$$

$$= ₹ 2,200$$

For 2nd year:

$$P = ₹ 2,200$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{2,200 \times 10 \times 1}{100} \right)$$

$$= ₹ \frac{22,000}{100}$$

$$= ₹ 220$$

And

$$\text{Final amount} = P + \text{Interest}$$

$$= ₹ 2,200 + 220$$

$$= ₹ 2,420$$

Hence, option 3 is the correct option.

Question 1(iii)

The difference between C.I. and S.I. on ₹ 6,000 at 8% per annum and in one year is:

1. ₹ 480

2. nothing

3. ₹ 240

4. none of these

Answer

$$P = ₹ 6,000$$

$$R = 8\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{6,000 \times 8 \times 1}{100} \right)$$

$$= ₹ \frac{48,000}{100}$$

$$= ₹ 480$$

And

$$\text{Amount} = P + \text{Interest}$$

$$= ₹ 6,000 + 480$$

$$= ₹ 6,480$$

And

Compound Interest = Final amount - Original Principal

$$= ₹6,480 - ₹6,000$$

$$= ₹480$$

Difference between S.I. and C.I. = ₹ 480 - ₹ 480 = 0.

Hence, option 2 is the correct option.

Question 1(iv)

The difference between the C.I. in 1 year and compound in interest in 2 years on ₹ 4,000 at 5% per annum is:

1. ₹ 10
2. ₹ 210
3. ₹ 200
4. ₹ 410

Answer

For 1st year:

$$P = ₹ 4,000$$

$$R = 5\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{4,000 \times 5 \times 1}{100} \right)$$

$$= ₹ \frac{20,000}{100}$$

$$= ₹200$$

And

$$\text{Amount} = P + \text{Interest}$$

$$= ₹4,000 + 200$$

$$= ₹4,200$$

For 2nd year:

$$P = ₹ 4,200$$

$$R = 5\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{4,200 \times 5 \times 1}{100} \right)$$

$$= ₹ \frac{21,000}{100}$$

$$= ₹210$$

And

$$\text{Final amount} = P + \text{Interest}$$

$$= ₹4,200 + 210$$

$$= ₹4,410$$

And

$$\text{Compound Interest} = \text{Final amount} - \text{Original Principal}$$

$$= ₹4,410 - ₹4,000$$

$$= ₹410$$

$$\begin{aligned} \text{Difference between the C.I. in 1 year and compound interest in 2 years} &= ₹ \\ 410 - ₹ 200 &= ₹ 210 \end{aligned}$$

Hence, option 2 is the correct option.

Question 2

A sum of ₹ 8,000 is invested for 2 years at 10% per annum compound interest.
Calculate:

(i) interest for the first year.

(ii) principal for the second year.

(iii) interest for the second year.

(iv) final amount at the end of the second year.

(v) compound interest earned in 2 years.

Answer

(i) **For 1st year:**

$$P = ₹ 8,000$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{8,000 \times 10 \times 1}{100} \right)$$

$$= ₹ \frac{80,000}{100}$$

$$= ₹ 800$$

Hence, interest for first year = ₹ 800.

(ii)

$$\text{Amount} = P + \text{Interest}$$

$$= ₹ 8,000 + 800$$

$$= ₹ 8,800$$

So, principal for the second year = ₹ 8,800.

(iii) **For 2nd year:**

$$P = ₹ 8,800$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{8,800 \times 10 \times 1}{100} \right)$$

$$= ₹ \frac{88,000}{100}$$

$$= ₹ 880$$

Hence, interest for the second year = ₹ 880.

(iv)

$$\text{Final amount} = P + \text{Interest}$$

$$= ₹ 8,800 + 880$$

$$= ₹ 9,680$$

So, final amount at the end of the second year = ₹ 9,680

(v)

Compound Interest = Final amount - Original Principal

$$= ₹9,680 - ₹8,000$$

$$= ₹1,680$$

Hence, compound interest earned in 2 years = ₹ 1,680.

Question 3

A man borrowed ₹ 20,000 for 2 years at 8% per year compound interest.

Calculate:

- (i) the interest for the first year.
- (ii) the interest for the second year.
- (iii) the final amount at the end of the second year.
- (iv) the compound interest for two years.

Answer

(i) For 1st year:

$$P = ₹ 20,000$$

$$R = 8\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{20,000 \times 8 \times 1}{100} \right)$$

$$= ₹ \frac{1,60,000}{100}$$

$$= ₹1,600$$

Hence, interest for first year = ₹ 1,600.

(ii)

$$\text{Amount} = P + \text{Interest}$$

$$= ₹20,000 + 1,600$$

$$= ₹21,600$$

For 2nd year:

$$P = ₹ 21,600$$

$$R = 8\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{21,600 \times 8 \times 1}{100} \right)$$

$$= ₹ \frac{1,72,800}{100}$$

$$= ₹1,728$$

Hence, interest for the second year = ₹ 1,728.

(iii)

$$\text{Final amount} = P + \text{Interest}$$

$$= ₹21,600 + 1,728$$

$$= ₹23,328$$

So, final amount at the end of the second year = ₹ 23,328.

(iv)

$$\text{Compound Interest} = \text{Final amount} - \text{Original Principal}$$

$$= ₹23,328 - ₹20,000$$

$$= ₹3,328$$

Hence, compound interest for 2 years = ₹ 3,328.

Question 4

Calculate the amount and the compound interest on ₹ 12,000 in 2 years at 10% per year.

Answer

Given:

$$P = ₹ 12,000$$

$$R = 10\%$$

$$n = 2 \text{ years}$$

$$\begin{aligned}A &= P \left[1 + \frac{R}{100} \right]^n \\&= 12,000 \left[1 + \frac{10}{100} \right]^2 \\&= 12,000 \left[1 + \frac{1}{10} \right]^2 \\&= 12,000 \left[\frac{10}{10} + \frac{1}{10} \right]^2 \\&= 12,000 \left[\frac{(10+1)}{10} \right]^2 \\&= 12,000 \left[\frac{11}{10} \right]^2 \\&= 12,000 \left[\frac{121}{100} \right] \\&= \left[\frac{1,45,200}{100} \right] \\&= 14,520\end{aligned}$$

Also

Compound Interest = Final amount - Original Principal

$$= ₹14,520 - ₹12,000$$

$$= ₹2,520$$

Hence, amount = ₹ 14,520 compound interest = ₹ 2,520.

Question 5

Calculate the amount and the compound interest on ₹ 10,000 in 3 years at 8% per annum.

Answer**Given:**

$$P = ₹ 10,000$$

$$R = 8\%$$

$$n = 3 \text{ years}$$

$$\begin{aligned} A &= P \left[1 + \frac{R}{100} \right]^n \\ &= 10,000 \left[1 + \frac{8}{100} \right]^3 \\ &= 10,000 \left[1 + \frac{2}{25} \right]^3 \\ &= 10,000 \left[\frac{25}{25} + \frac{2}{25} \right]^3 \\ &= 10,000 \left[\frac{(25 + 2)}{25} \right]^3 \\ &= 10,000 \left[\frac{27}{25} \right]^3 \\ &= 10,000 \left[\frac{19683}{15625} \right] \end{aligned}$$

$$= \left[\frac{19,68,30,000}{15625} \right]$$

$$= 12,597.12$$

Also

Compound Interest = Final amount - Original Principal

$$= ₹12,597.12 - ₹10,000$$

$$= ₹2,597.12$$

Hence, amount = ₹ 12,597.12 compound interest = ₹ 2,597.12.

Question 6

Calculate the compound interest on ₹ 5,000 in 2 years, if the rates of interest for successive years are 10% and 12% respectively.

Answer

Given:

$$P = ₹ 5,000$$

$$T = 2 \text{ years}$$

$$R_1 = 10\%$$

$$R_2 = 12\%$$

As we know,

$$\begin{aligned}A &= P \left[1 + \frac{R_1}{100} \right] \left[1 + \frac{R_2}{100} \right] \\&= 5,000 \left[1 + \frac{10}{100} \right] \left[1 + \frac{12}{100} \right] \\&= 5,000 \left[1 + \frac{1}{10} \right] \left[1 + \frac{3}{25} \right] \\&= 5,000 \left[\frac{10}{10} + \frac{1}{10} \right] \left[\frac{25}{25} + \frac{3}{25} \right] \\&= 5,000 \left[\frac{(10+1)}{10} \right] \left[\frac{(25+3)}{25} \right] \\&= 5,000 \left[\frac{11}{10} \right] \left[\frac{28}{25} \right] \\&= \left[\frac{15,40,000}{250} \right] \\&= ₹6,160\end{aligned}$$

$$\text{C.I.} = A - P$$

$$= ₹6,160 - ₹5,000$$

$$= ₹1,160$$

Hence, compound interest = ₹ 1,160

Question 7

Calculate the compound interest on ₹ 15,000 in 3 years; if the rates of interest for successive years are 6%, 8% and 10% respectively.

Answer**Given:**

$$P = ₹ 15,000$$

$$T = 3 \text{ years}$$

$$R_1 = 6\%$$

$$R_2 = 8\%$$

$$R_3 = 10\%$$

As we know,

$$\begin{aligned} A &= P \left[1 + \frac{R_1}{100} \right] \left[1 + \frac{R_2}{100} \right] \left[1 + \frac{R_3}{100} \right] \\ &= 15,000 \left[1 + \frac{6}{100} \right] \left[1 + \frac{8}{100} \right] \left[1 + \frac{10}{100} \right] \\ &= 15,000 \left[1 + \frac{3}{50} \right] \left[1 + \frac{2}{25} \right] \left[1 + \frac{1}{10} \right] \\ &= 15,000 \left[\frac{50}{50} + \frac{3}{50} \right] \left[\frac{25}{25} + \frac{2}{25} \right] \left[\frac{10}{10} + \frac{1}{10} \right] \\ &= 15,000 \left[\frac{(50 + 3)}{50} \right] \left[\frac{(25 + 2)}{25} \right] \left[\frac{(10 + 1)}{10} \right] \\ &= 15,000 \left[\frac{53}{50} \right] \left[\frac{27}{25} \right] \left[\frac{11}{10} \right] \end{aligned}$$

$$= \left[\frac{23,61,15,000}{12500} \right]$$

$$= 18,889.20$$

$$\text{C.I.} = A - P$$

$$= ₹18,889.20 - ₹15,000$$

$$= ₹3,889.20$$

Hence, compound interest = ₹ 3,889.20

Question 8

Mohan borrowed ₹ 16,000 for 3 years at 5% per annum compound interest. Calculate the amount that Mohan would have to pay at the end of 3 years.

Answer

Given:

$$P = ₹ 16,000$$

$$n = 3 \text{ years}$$

$$R = 5\%$$

As we know,

$$\begin{aligned} A &= P \left[1 + \frac{R}{100} \right]^n \\ &= 16,000 \left[1 + \frac{5}{100} \right]^3 \end{aligned}$$

$$\begin{aligned} &= 16,000 \left[1 + \frac{1}{20} \right]^3 \\ &= 16,000 \left[\frac{20}{20} + \frac{1}{20} \right]^3 \\ &= 16,000 \left[\frac{(20 + 1)}{20} \right]^3 \\ &= 16,000 \left[\frac{21}{20} \right]^3 \\ &= 16,000 \left[\frac{9261}{8000} \right] \\ &= \left[\frac{14,81,76,000}{8000} \right] \\ &= 18,522 \end{aligned}$$

Hence, amount = ₹ 18,522

Question 9

Rekha borrowed ₹ 40,000 for 3 years at 10% per annum compound interest. Calculate the interest paid by her for the second year.

Answer

For 1st year:

$$P = ₹ 40,000$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\begin{aligned}\text{Interest} &= \left(\frac{P \times R \times T}{100} \right) \\ &= ₹ \left(\frac{40,000 \times 10 \times 1}{100} \right) \\ &= ₹ \frac{4,00,000}{100} \\ &= ₹4,000\end{aligned}$$

And

$$\begin{aligned}\text{Amount} &= P + \text{Interest} \\ &= ₹40,000 + 4,000 \\ &= ₹44,000\end{aligned}$$

For 2nd year:

$$P = ₹ 44,000$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\begin{aligned}\text{Interest} &= \left(\frac{P \times R \times T}{100} \right) \\ &= ₹ \left(\frac{44,000 \times 10 \times 1}{100} \right)\end{aligned}$$

$$= ₹ \frac{4,40,000}{100}$$

$$= ₹4,400$$

Hence, compound interest = ₹ 4,400.

Question 10

Calculate the compound interest for the second year on ₹ 15,000 invested for 5 years at 6% per annum.

Answer

For 1st year:

$$P = ₹ 15,000$$

$$R = 6\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{15,000 \times 6 \times 1}{100} \right)$$

$$= ₹ \frac{90,000}{100}$$

$$= ₹900$$

And

$$\text{Amount} = P + \text{Interest}$$

$$= ₹15,000 + 900$$

$$= ₹15,900$$

For 2nd year:

$$P = ₹ 15,900$$

$$R = 6\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{15,900 \times 6 \times 1}{100} \right)$$

$$= ₹ \frac{954,000}{100}$$

$$= ₹954$$

Hence, compound interest = ₹ 954.

Question 11

A man invests ₹ 9,600 at 10% per annum compound interest for 3 years.

Calculate:

(i) the interest for the first year.

(ii) the amount at the end of the first year.

(iii) the interest for the second year.

(iv) the interest for the third year.

Answer

(i) **For 1st year:**

$$P = ₹ 9,600$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{9,600 \times 10 \times 1}{100} \right)$$

$$= ₹ \frac{96,000}{100}$$

$$= ₹ 960$$

Hence, interest for first year = ₹ 960.

(ii)

$$\text{Amount} = P + \text{Interest}$$

$$= ₹ 9,600 + 960$$

$$= ₹ 10,560$$

So, amount at the end of the first year = ₹ 10,560.

(iii) **For 2nd year:**

$$P = ₹ 10,560$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{10,560 \times 10 \times 1}{100} \right)$$

$$= ₹ \frac{1,05,600}{100}$$

$$= ₹ 1,056$$

Hence, interest for the second year = ₹ 1,056.

(iv)

Amount at end of 2nd year = P + Interest

$$= ₹ 10,560 + 1,056$$

$$= ₹ 11,616$$

For 3rd year:

$$P = ₹ 11,616$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\begin{aligned}\text{Interest} &= \left(\frac{P \times R \times T}{100} \right) \\ &= ₹ \left(\frac{11,616 \times 10 \times 1}{100} \right) \\ &= ₹ \frac{1,16,160}{100} \\ &= ₹1,161.60\end{aligned}$$

Hence, interest for the third year = ₹ 1,161.60.

Question 12

A person invests ₹ 5,000 for two years at a certain rate of interest compounded annually. At the end of one year, this sum amounts to ₹ 5,600. Calculate:

- (i) the rate of interest per annum.
- (ii) the amount at the end of the second year.

Answer

(i) For 1st year:

$$P = ₹ 5,000$$

$$T = 1 \text{ year}$$

$$A = ₹ 5,600$$

As we know,

$$A = P + S.I.$$

$$\Rightarrow 5,600 = 5,000 + S.I.$$

$$\Rightarrow S.I. = 5,600 - 5,000$$

$$\Rightarrow S.I. = 600$$

Let the rate be r

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 600 = ₹ \left(\frac{5,000 \times r \times 1}{100} \right)$$

$$\Rightarrow 600 = ₹ \left(\frac{5,000r}{100} \right)$$

$$\Rightarrow 600 = 50r$$

$$\Rightarrow r = \frac{600}{50}$$

$$\Rightarrow r = 12$$

Hence, rate of interest = 12%.

(ii) For 2nd year:

$$P = ₹ 5,600$$

$$R = 12\%$$

$$T = 1 \text{ year}$$

$$\begin{aligned}\text{Interest} &= \left(\frac{P \times R \times T}{100} \right) \\ &= ₹ \left(\frac{5,600 \times 12 \times 1}{100} \right) \\ &= ₹ \frac{67,200}{100} \\ &= ₹672\end{aligned}$$

$$\text{Final amount} = P + \text{Interest}$$

$$\begin{aligned}&= ₹(5,600 + 672) \\ &= ₹6,272\end{aligned}$$

Question 13

Calculate the difference between the compound interest and the simple interest on ₹ 7,500 in two years and at 8% per annum.

Answer

$$P = ₹ 7,500$$

$$R = 8\%$$

$$T = 2 \text{ years}$$

For simple interest

$$\begin{aligned}\text{Interest} &= \left(\frac{P \times R \times T}{100} \right) \\ &= \left(\frac{7,500 \times 8 \times 2}{100} \right) \\ &= \left(\frac{1,20,000}{100} \right) \\ &= ₹1,200\end{aligned}$$

For compound interest

$$\begin{aligned}A &= P \left[1 + \frac{R}{100} \right]^n \\ &= 7,500 \left[1 + \frac{8}{100} \right]^2 \\ &= 7,500 \left[1 + \frac{2}{25} \right]^2 \\ &= 7,500 \left[\frac{25}{25} + \frac{2}{25} \right]^2 \\ &= 7,500 \left[\frac{(25 + 2)}{25} \right]^2 \\ &= 7,500 \left[\frac{27}{25} \right]^2 \\ &= 7,500 \left[\frac{729}{625} \right]^2\end{aligned}$$

$$= \left[\frac{54,67,500}{625} \right]^2$$

$$= ₹8,748$$

And

$$\text{C.I.} = A - P$$

$$= ₹(8,748 - 7,500)$$

$$= ₹1,248$$

Difference between C.I. and S.I.

$$= ₹(1,248 - 1,200)$$

$$= ₹48$$

The difference between C.I. and S.I. = ₹48.

Question 14

Calculate the difference between the compound interest and the simple interest on ₹8,000 in three years at 10% per annum.

Answer

$$P = ₹8,000$$

$$R = 10\%$$

$$T = 3 \text{ years}$$

For simple interest

$$\begin{aligned}\text{Interest} &= \left(\frac{P \times R \times T}{100} \right) \\ &= \left(\frac{8,000 \times 10 \times 3}{100} \right) \\ &= \left(\frac{2,40,000}{100} \right) \\ &= ₹2,400\end{aligned}$$

For compound interest

$$\begin{aligned}A &= P \left[1 + \frac{R}{100} \right]^n \\ &= 8,000 \left[1 + \frac{10}{100} \right]^3 \\ &= 8,000 \left[1 + \frac{1}{10} \right]^3 \\ &= 8,000 \left[\frac{10}{10} + \frac{1}{10} \right]^3 \\ &= 8,000 \left[\frac{(10 + 1)}{10} \right]^3 \\ &= 8,000 \left[\frac{11}{10} \right]^3 \\ &= 8,000 \left[\frac{1,331}{1000} \right]\end{aligned}$$

$$= \left[\frac{10,64,80,000}{1000} \right]$$

$$= ₹10,648$$

And

$$\text{C.I.} = A - P$$

$$= ₹(10,648 - 8,000)$$

$$= ₹2,648$$

Difference between C.I. and S.I.

$$= ₹(2,648 - 2,400)$$

$$= ₹248$$

The difference between C.I. and S.I. = ₹248.

Question 15

Rohit borrowed ₹40,000 for 2 years at 10% per annum C.I. and Manish borrowed the same sum for the same time at 10.5% per annum simple interest. Which of these two gives less interest and by how much?

Answer

For simple interest

$$P = ₹40,000$$

$$R = 10.5\%$$

$$T = 2 \text{ years}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= \left(\frac{40,000 \times 10.5 \times 2}{100} \right)$$

$$= \left(\frac{8,40,000}{100} \right)$$

$$= ₹8,400$$

For compound interest

$$P = ₹40,000$$

$$R = 10\%$$

$$n = 2 \text{ years}$$

$$A = P \left[1 + \frac{R}{100} \right]^n$$

$$= 40,000 \left[1 + \frac{10}{100} \right]^2$$

$$= 40,000 \left[1 + \frac{1}{10} \right]^2$$

$$= 40,000 \left[\frac{10}{10} + \frac{1}{10} \right]^2$$

$$= 40,000 \left[\frac{(10 + 1)}{10} \right]^2$$

$$= 40,000 \left[\frac{11}{10} \right]^2$$

$$= 40,000 \left[\frac{121}{100} \right]$$

$$= \left[\frac{48,40,000}{100} \right]$$

$$= ₹48,400$$

And

$$\text{C.I.} = A - P$$

$$= ₹(48,400 - 40,000)$$

$$= ₹8,400$$

∴ C.I. is equals to S.I.

Hence, Both give equal interest.

Question 16

Mr. Sharma lends ₹ 24,000 at 13% p.a. simple interest and an equal sum at 12% p.a. compound interest. Find the total interest earned by Mr. Sharma in 2 years.

Answer

For simple interest

$$P = ₹ 24,000$$

$$R = 13\%$$

$$T = 2 \text{ years}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= \left(\frac{24,000 \times 13 \times 2}{100} \right)$$

$$= \left(\frac{6,24,000}{100} \right)$$

$$= ₹6,240$$

For compound interest

$$P = ₹ 24,000$$

$$R = 12\%$$

$$n = 2 \text{ years}$$

$$A = P \left[1 + \frac{R}{100} \right]^n$$

$$= 24,000 \left[1 + \frac{12}{100} \right]^2$$

$$= 24,000 \left[1 + \frac{3}{25} \right]^2$$

$$= 24,000 \left[\frac{25}{25} + \frac{3}{25} \right]^2$$

$$= 24,000 \left[\frac{(25 + 3)}{25} \right]^2$$

$$\begin{aligned} &= 24,000 \left[\frac{28}{25} \right]^2 \\ &= 24,000 \left[\frac{784}{625} \right] \\ &= \left[\frac{1,88,16,000}{625} \right] \\ &= ₹30,105.60 \end{aligned}$$

And

$$\text{C.I.} = A - P$$

$$= ₹(30,105.6 - 24,000)$$

$$= ₹6,105.60$$

$$\text{Total interest} = ₹ 6,240 + ₹ 6,105.60 = ₹ 12,345.60$$

Hence, total interest earned by Mr. Sharma in 2 years = ₹ 12,345.60

Question 17

Peter borrows ₹ 12,000 for 2 years at 10% p.a. compound interest. He repays ₹ 8,000 at the end of first year. Find:

- (i) the amount at the end of first year, before making the repayment.
- (ii) the amount at the end of first year, after making the repayment.
- (iii) the principal for the second year.
- (iv) the amount to be paid at the end of the second year to clear the account.

Answer**For 1st year:**

$$P = ₹ 12,000$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{12,000 \times 10 \times 1}{100} \right)$$

$$= ₹ \frac{1,20,000}{100}$$

$$= ₹ 1,200$$

And

$$\text{Amount} = P + \text{Interest}$$

$$= ₹ 12,000 + 1,200$$

$$= ₹ 13,200$$

Hence, the amount at the end of first year, before making the repayment = ₹ 13,200

(ii) The amount at the end of first year, after making the repayment = ₹ 13,200 - ₹ 8,000

$$= ₹ 5,200$$

Hence, the amount at the end of first year, after making the repayment = ₹ 5,200

(iii) So, the principal amount for second year = ₹ 5,200

(iv) For 2nd year:

$$P = ₹ 5,200$$

$$R = 10\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{5,200 \times 10 \times 1}{100} \right)$$

$$= ₹ \frac{52,000}{100}$$

$$= ₹ 520$$

And

$$\text{Final amount} = P + \text{Interest}$$

$$= ₹ 5,200 + 520$$

$$= ₹ 5,720$$

Hence, the amount to be paid at the end of the second year to clear the account = ₹ 5,720.

Question 18

Gautam takes a loan of ₹ 16,000 for 2 years at 15% p.a. compound interest. He repays ₹ 9,000 at the end of first year. How much must he pay at the end of second year to clear the debt?

Answer

For 1st year:

$$P = ₹ 16,000$$

$$R = 15\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{16,000 \times 15 \times 1}{100} \right)$$

$$= ₹ \frac{2,40,000}{100}$$

$$= ₹ 2,400$$

And

$$\text{Amount} = P + \text{Interest}$$

$$= ₹ 16,000 + 2,400$$

$$= ₹18,400$$

The amount at the end of first year, after making the repayment = ₹ 18,400 - ₹ 9,000

$$= ₹ 9,400$$

Hence, the amount at the end of first year, after making the repayment = ₹ 9,400

For 2nd year:

$$P = ₹ 9,400$$

$$R = 15\%$$

$$T = 1 \text{ year}$$

$$\text{Interest} = \left(\frac{P \times R \times T}{100} \right)$$

$$= ₹ \left(\frac{9,400 \times 15 \times 1}{100} \right)$$

$$= ₹ \frac{141,000}{100}$$

$$= ₹1,410$$

And

$$\text{Final amount} = P + \text{Interest}$$

$$= ₹9,400 + 1,410$$

$$= ₹10,810$$

Hence, the amount to be paid at the end of the second year to clear the account = ₹ 10,810.

Question 19

A certain sum of money, invested for 5 years at 8% p.a. simple interest, earns an interest of ₹ 12,000. Find:

(i) the sum of money.

(ii) the compound interest earned by this money in two years at 10% p.a. compound interest.

Answer

(i) **For simple interest**

$$T = 5 \text{ years}$$

$$R = 8\%$$

$$\text{S.I.} = ₹ 12,000$$

Let the principal amount be P .

As we know,

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 12,000 = \left(\frac{P \times 8 \times 5}{100} \right)$$

$$\Rightarrow 12,000 = \left(\frac{40P}{100}\right)$$

$$\Rightarrow 12,000 = \left(\frac{2P}{5}\right)$$

$$\Rightarrow 12,000 = \frac{2P}{5}$$

$$\Rightarrow P = \frac{5 \times 12,000}{2}$$

$$\Rightarrow P = \frac{60,000}{2}$$

$$\Rightarrow P = 30,000$$

Hence, the principal amount = ₹ 30,000.

(ii) For compound interest

$$P = ₹ 30,000$$

$$R = 10\%$$

$$n = 2 \text{ years}$$

$$A = P \left[1 + \frac{R}{100}\right]^n$$

$$\Rightarrow A = 30,000 \left[1 + \frac{10}{100}\right]^2$$

$$= 30,000 \left[1 + \frac{1}{10}\right]^2$$

$$= 30,000 \left[\frac{10}{10} + \frac{1}{10} \right]^2$$

$$= 30,000 \left[\frac{(10 + 1)}{10} \right]^2$$

$$= 30,000 \left[\frac{11}{10} \right]^2$$

$$= 30,000 \left[\frac{121}{100} \right]$$

$$= \left[\frac{36,30,000}{100} \right]$$

$$= 36,300$$

And

$$C.I. = A - P$$

$$\Rightarrow C.I. = 36,300 - 30,000$$

$$= 6,300$$

Hence, the compound interest ₹ 6,300

Question 20

Find the amount and the C.I. on ₹ 12,000 in one year at 10% per annum compounded half-yearly.

Answer

Given:

$$P = ₹ 12,000$$

$$R = 10\%$$

$$n = 1 \text{ years}$$

When the interest is compounded half-yearly:

$$A = P \left[1 + \frac{R}{2 \times 100} \right]^{2 \times n}$$

$$= 12,000 \left[1 + \frac{10}{200} \right]^{2 \times 1}$$

$$= 12,000 \left[1 + \frac{1}{20} \right]^2$$

$$= 12,000 \left[\frac{20}{20} + \frac{1}{20} \right]^2$$

$$= 12,000 \left[\frac{(20 + 1)}{20} \right]^2$$

$$= 12,000 \left[\frac{21}{20} \right]^2$$

$$= 12,000 \left[\frac{441}{400} \right]$$

$$= \left[\frac{52,92,000}{400} \right]$$

$$= 13,230$$

And

$$\text{C.I.} = A - P$$

$$\Rightarrow \text{C.I.} = 13,230 - 12,000$$

$$= 1,230$$

Hence, the amount = ₹ 13,230 and the compound interest ₹ 1,230

Question 21

Find the amount and the C.I. on ₹ 8,000 in $1\frac{1}{2}$ years at 20% per year compounded half-yearly.

Answer

Given:

$$P = ₹ 8,000$$

$$R = 20\%$$

$$n = 1\frac{1}{2} \text{ years}$$

$$= \frac{3}{2} \text{ years}$$

When the interest is compounded half-yearly:

$$A = P \left[1 + \frac{R}{2 \times 100} \right]^{2 \times n}$$

$$= 8,000 \left[1 + \frac{20}{200} \right]^{2 \times \frac{3}{2}}$$

$$\begin{aligned} &= 8,000 \left[1 + \frac{1}{10} \right]^3 \\ &= 8,000 \left[\frac{10}{10} + \frac{1}{10} \right]^3 \\ &= 8,000 \left[\frac{(10 + 1)}{10} \right]^3 \\ &= 8,000 \left[\frac{11}{10} \right]^3 \\ &= 8,000 \left[\frac{1331}{1000} \right] \\ &= \left[\frac{1,06,48,000}{1000} \right] \\ &= 10,648 \end{aligned}$$

And

$$\text{C.I.} = A - P$$

$$\Rightarrow \text{C.I.} = 10,648 - 8,000$$

$$= 2,648$$

Hence, the amount = ₹ 10,648 and the compound interest = ₹ 2,648

Question 22

Find the amount and the compound interest on ₹ 24,000 for 2 years at 10% per annum compounded yearly.

Answer

Given:

$$P = ₹ 24,000$$

$$R = 10\%$$

$$n = 2 \text{ years}$$

$$A = P \left[1 + \frac{R}{100} \right]^n$$

$$\Rightarrow A = 24,000 \left[1 + \frac{10}{100} \right]^2$$

$$= 24,000 \left[1 + \frac{1}{10} \right]^2$$

$$= 24,000 \left[\frac{10}{10} + \frac{1}{10} \right]^2$$

$$= 24,000 \left[\frac{(10 + 1)}{10} \right]^2$$

$$= 24,000 \left[\frac{11}{10} \right]^2$$

$$= 24,000 \left[\frac{121}{100} \right]$$

$$= \left[\frac{29,04,000}{100} \right]$$

$$= 29,040$$

And

$$\text{C.I.} = A - P$$

$$\Rightarrow \text{C.I.} = 29,040 - 24,000$$

$$= 5,040$$

Hence, the amount = ₹ 29,040 and the compound interest = ₹ 5,040

Test Yourself

Question 1(i)

The S.I. on a certain sum in 3 years and at 8% per year is ₹ 720. The sum is:

1. ₹ 6,000

2. ₹ 9,000

3. ₹ 3,000

4. ₹ 4,000

Answer

Given:

$$T = 3 \text{ years}$$

$$R = 8\%$$

$$\text{S.I.} = ₹ 720$$

Let the principal be P .

As we know,

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 720 = \left(\frac{P \times 8 \times 3}{100} \right)$$

$$\Rightarrow 720 = \left(\frac{24P}{100} \right)$$

$$\Rightarrow 720 = \left(\frac{6P}{25} \right)$$

$$\Rightarrow P = \left(\frac{25 \times 720}{6} \right)$$

$$\Rightarrow P = \left(\frac{18,000}{6} \right)$$

$$\Rightarrow P = 3,000$$

Hence, option 3 is the correct option.

Question 1(ii)

A sum of money becomes $\frac{5}{4}$ of itself in 5 years. The rate of interest is:

1. 10%

2. 5%

3. 8%

4. 15%

Answer

Given:

$T = 5$ years

$A = ₹ \frac{5}{4}$ of P

Let the principal be P and the rate be r .

As we know,

$$A = P + \text{S.I.}$$

$$\Rightarrow \frac{5}{4}P = P + \text{S.I.}$$

$$\Rightarrow \text{S.I.} = \frac{5}{4}P - P$$

$$\Rightarrow \text{S.I.} = \frac{5}{4}P - \frac{4}{4}P$$

$$\Rightarrow \text{S.I.} = \frac{(5 - 4)}{4} P$$

$$\Rightarrow \text{S.I.} = \frac{1}{4} P$$

And

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow \frac{1}{4} \times P = \left(\frac{P \times r \times 5}{100} \right)$$

$$\Rightarrow \frac{1}{4} P = \left(\frac{5Pr}{100} \right)$$

$$\Rightarrow \frac{1}{4} \cancel{P} = \left(\frac{5r\cancel{P}}{100} \right)$$

$$\Rightarrow \frac{1}{4} = \left(\frac{5r}{100} \right)$$

$$\Rightarrow \frac{1}{4} = \left(\frac{r}{20} \right)$$

$$\Rightarrow r = \left(\frac{1 \times 20}{4} \right)$$

$$\Rightarrow r = \left(\frac{20}{4} \right)$$

$$\Rightarrow r = 5$$

Hence, option 2 is the correct option.

Question 1(iii)

$\frac{3}{5}$ part of certain sum is lent at S.I. and the remaining is lent at C.I. If the rate of interest in both the cases is 20%. On the whole the total interest in 1 year is ₹ 1,000 then the sum is:

1. ₹ 2,000
2. ₹ 4,000
3. ₹ 2,500
4. ₹ 5,000

Answer

Let the principal amount be ₹ x .

Let the first part be ₹ $\frac{3}{5}$ of x and the second part be ₹ $x - \frac{3}{5}x = \frac{5}{5}x - \frac{3}{5}x = \frac{2}{5}x$.

Hence

$$P_1 = ₹ \frac{3x}{5}$$

$$R_1 = 20\%$$

$$T_1 = 1 \text{ year}$$

$$\begin{aligned} \text{S.I.} &= \left(\frac{P \times R \times T}{100} \right) \\ &= \left(\frac{3x \times 20 \times 1}{5 \times 100} \right) \end{aligned}$$

$$= \frac{60x}{500}$$

$$= \frac{3x}{25}$$

$$P_2 = ₹ \frac{2x}{5}$$

$$R_2 = 20$$

$$T_2 = 1 \text{ year}$$

$$A = P \left[1 + \frac{R}{100} \right]^n$$

$$= \frac{2x}{5} \left[1 + \frac{20}{100} \right]^1$$

$$= \frac{2x}{5} \left[1 + \frac{1}{5} \right]$$

$$= \frac{2x}{5} \left[\frac{5}{5} + \frac{1}{5} \right]$$

$$= \frac{2x}{5} \left[\frac{(5+1)}{5} \right]$$

$$= \frac{2x}{5} \left[\frac{6}{5} \right]$$

$$= \frac{12x}{25}$$

And

$$\text{C.I.} = A - P$$

$$= \frac{12x}{25} - \frac{2x}{5}$$

$$= \frac{12x}{25} - \frac{5 \times 2x}{5 \times 5}$$

$$= \frac{12x}{25} - \frac{10x}{25}$$

$$= \frac{(12 - 10)x}{25}$$

$$= \frac{2x}{25}$$

Total amount = ₹ 1,000

$$\frac{3x}{25} + \frac{2x}{25} = 1,000$$

$$\Rightarrow \frac{(3x + 2x)}{25} = 1,000$$

$$\Rightarrow \frac{5x}{25} = 1,000$$

$$\Rightarrow \frac{x}{5} = 1,000$$

$$\Rightarrow x = 1,000 \times 5$$

$$\Rightarrow x = 5,000$$

Hence, option 4 is the correct option.

Question 1(iv)

The amount, of ₹ 1,000 invested for 2 years at 5% per annum compounded annually is:

1. ₹ 1,100
2. ₹ 1,102.50
3. ₹ 1,200
4. ₹ 8,000

Answer**Given:**

$$P = ₹ 1,000$$

$$R = 5\%$$

$$n = 2 \text{ years}$$

$$\begin{aligned} A &= P \left[1 + \frac{R}{100} \right]^n \\ &= 1,000 \left[1 + \frac{5}{100} \right]^2 \\ &= 1,000 \left[1 + \frac{1}{20} \right]^2 \\ &= 1,000 \left[\frac{20}{20} + \frac{1}{20} \right]^2 \\ &= 1,000 \left[\frac{(20 + 1)}{20} \right]^2 \end{aligned}$$

$$= 1,000 \left[\frac{21}{20} \right]^3$$

$$= 1,000 \left[\frac{441}{400} \right]$$

$$= \left[\frac{4,41,000}{400} \right]$$

$$= 1,102.50$$

Hence, option 2 is the correct option.

Question 1(v)

If the interest is compounded half-yearly, the time is:

1. halved
2. doubled
3. tripled
4. not changed

Answer

When the interest is compounded half-yearly:

$$A = P \left[1 + \frac{R}{2 \times 100} \right]^{2 \times n}$$

2 x n means time is doubled.

Hence, option 2 is the correct option.

Question 1(vi)

Statement 1: On a certain sum, at the same rate of interest and for the same time period, compound interest is always greater than the simple interest.

Statement 2: In compound interest, the principal remains constant for the whole time period, however the compound interest keeps increasing every year.

Which of the following options is correct?

1. Both the statements are true.
2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

On a certain sum, at the same rate of interest and for the same time period,

For a time period of one year, compound interest (CI) = simple interest (SI).

But for more than one year, CI is always greater than SI, because interest is added to the principal after each compounding period, increasing the amount on which interest is calculated.

On a certain sum, at the same rate of interest and for the same time period, compound interest is always greater or equal to the simple interest.

So, statement 1 is false.

In compound interest, the principal does not remain constant — it increases after each compounding period because interest is added to it.

So, statement 2 is false.

Hence, option 2 is the correct option.

Question 1(vii)

Assertion (A) : The simple interest on a certain sum is $\frac{9}{16}$ of the principal. If the number representing the rate of interest in percent and time in years are equal, then the time for which the principle is lent out is $7\frac{1}{2}$ years.

Reason (R) : In simple interest,

$$\text{Time} = \frac{\text{S.I.} \times 100}{\text{Principal} \times \text{Rate}}.$$

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Let the principal be ₹ P.

$$\text{S.I.} = \frac{9}{16} \times P$$

Rate of interest = a%

Time = a years

By formula,

$$\text{S.I.} = \frac{P \times R \times T}{100}$$

Substituting values we get :

$$\Rightarrow \frac{9}{16}P = \frac{P \times a \times a}{100}$$

$$\Rightarrow \frac{9}{16}\cancel{P} = \frac{\cancel{P} \times a \times a}{100}$$

$$\Rightarrow \frac{9}{16} = \frac{a \times a}{100}$$

$$\Rightarrow \frac{9}{16} = \frac{a^2}{100}$$

$$\Rightarrow a^2 = \frac{9 \times 100}{16}$$

$$\Rightarrow a^2 = \frac{900}{16}$$

$$\Rightarrow a = \sqrt{\frac{900}{16}}$$

$$\Rightarrow a = \frac{30}{4}$$

$$\Rightarrow a = \frac{15}{2}$$

$$\Rightarrow a = 7\frac{1}{2}$$

$$\text{Time} = 7\frac{1}{2} \text{ years.}$$

So, assertion (A) is true.

By formula;

$$\Rightarrow S.I. = \frac{P \times R \times T}{100}$$

$$\Rightarrow T = \frac{S.I. \times 100}{P \times R}$$

$$\Rightarrow \text{Time} = \frac{\text{Simple Interest} \times 100}{\text{Principal} \times \text{Rate}}$$

So, reason (R) is true and reason (R) clearly explains assertion (A).

Hence, option 1 is the correct option.

Question 1(viii)

Assertion (A) : The simple interest on ₹ 15,000 in 2 years at 6% p.a. is ₹ 1,800.

Then compound interest on the same sum at the same rate of interest for 2 years will never be less than ₹ 1,800.

Reason (R) : For a given principal, rate and time, both simple interest and compound interest are equal for the 1st year.

1. Both A and R are correct, and R is the correct explanation for A.

2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

We know that,

If time is greater than 1 year, then compound interest on the same sum at the same rate of interest for same time will always be greater than simple interest.

∴ If simple interest on ₹ 15,000 in 2 years at 6% p.a. is ₹ 1,800. Then compound interest on the same sum at the same rate of interest for 2 years will never be less than ₹ 1,800.

So, assertion (A) is true.

For a given principal, rate and time, both simple interest and compound interest are equal for the 1st year.

So, reason (R) is true and reason (R) does not explain assertion (A).

Hence, option 2 is the correct option.

Question 1(ix)

Assertion (A): Compound interest for the 2nd year on ₹ 8,000 at 5% p.a. is ₹ 820.

Reason (R): Compound interest for 2 years = Amount at the end of 2nd year - original sum.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given,

$$P = ₹ 8,000$$

$$R = 5\%$$

$$T = 2 \text{ years}$$

By formula,

$$\text{Amount} = P \left(1 + \frac{R}{100} \right)^n$$

$$\text{Amount after one year} = 8000 \left(1 + \frac{5}{100} \right)^1$$

$$= 8000(1 + 0.05)$$

$$= 8000 \times (1.05)$$

$$= ₹8,400.$$

Amount at the end of first year = ₹ 8,400

For 2nd year :

$$P = ₹ 8,400$$

$$R = 5\%$$

$$T = 1 \text{ year}$$

$$\text{Interest for 2nd year} = \frac{P \times R \times T}{100}$$

$$= \frac{8,400 \times 5 \times 1}{100}$$

$$= ₹420.$$

∴ Compound interest for the 2nd year on ₹ 8,000 at 5% p.a. is ₹ 420.

So, assertion (A) is false.

By formula,

Compound interest for 2 years = Amount at the end of 2nd year - original sum.

So, reason (R) is true.

Hence, option 4 is the correct option.

Question 1(x)

Assertion (A) : On ₹ 8,750 at 8% p.a., the simple interest for 1st year is equal to the simple interest for 4th year.

Reason (R) : Simple interest for 4th year = Amount at the end of 4th year - original sum.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

In simple interest, the interest is calculated only on the original principal and remains the same every year.

Thus, SI for the 1st year is equal to SI for the 4th year—in fact, it's equal for every year.

So, assertion (A) is true.

Amount at the end of 4th year - original sum = Total simple interest for 4 years

That formula gives the total simple interest for 4 years, not just for the 4th year.

So, reason (R) is false.

Hence, option 3 is the correct option.

Question 2

Mohan lends ₹ 4,800 to John for $4\frac{1}{2}$ years and ₹ 2,500 to Shyam for 6 years and receives a total sum of ₹ 2,196 as interest. Find the rate per cent per annum, provided it is the same in both the cases.

Answer

For Mohan,

$$P = ₹ 4,800$$

$$T = 4\frac{1}{2} \text{ years}$$

$$= \frac{9}{2} \text{ years}$$

Let the rate be r

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$= \left(\frac{4,800 \times r \times 9}{2 \times 100} \right)$$

$$= \left(\frac{43,200r}{200} \right)$$

$$= 216r$$

For Shyam,

$$P = ₹ 2,500$$

$$T = 6 \text{ years}$$

Let the rate be r

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$= \left(\frac{2,500 \times r \times 6}{100} \right)$$

$$= \left(\frac{15,000r}{100} \right)$$

$$= 150r$$

$$\text{Total interest} = ₹ 2,196$$

$$₹ 216r + ₹ 150r = ₹ 2,196$$

$$₹ 366r = ₹ 2,196$$

$$r = \frac{2196}{366}$$

$$r = 6$$

Hence, the rate per cent per annum = 6%.

Question 3

John lent ₹ 2,550 to Mohan at 7.5 percent per annum. If Mohan discharges the debt after 8 months by giving an old television and ₹ 1,422.50, find the price of the television.

Answer

Given:

$$P = ₹ 2,550$$

$$R = 7.5\%$$

$$T = 8 \text{ months} = \frac{8}{12} \text{ years}$$

$$= \frac{2}{3} \text{ years}$$

$$\begin{aligned}\text{S.I.} &= \left(\frac{P \times R \times T}{100} \right) \\ &= \left(\frac{2,550 \times 7.5 \times 2}{3 \times 100} \right) \\ &= \left(\frac{38,250}{300} \right) \\ &= 127.50\end{aligned}$$

And

$$A = P + \text{S.I.}$$

$$\Rightarrow A = 2,550 + 127.50$$

$$\Rightarrow A = 2,677.50$$

Mohan paid in cash = ₹ 1,422.50

Price of the television = Amount - Paid in cash

$$= ₹ 2,677.50 - ₹ 1,422.50$$

$$= ₹ 1,255$$

Hence, the cost of television = ₹ 1,255

Question 4

Divide ₹ 10,800 into two parts so that if one part is put at 18% per annum S.I. and the other part is put at 20% p.a. S.I. the total annual interest is ₹ 2,060.

Answer

Let the first part be ₹ x and the second part be ₹ $(10,800 - x)$.

Hence

$$P_1 = ₹ x$$

$$R_1 = 18\%$$

$$T_1 = 1 \text{ year}$$

$$\begin{aligned} \text{S.I.} &= \left(\frac{P \times R \times T}{100} \right) \\ &= \left(\frac{x \times 18 \times 1}{100} \right) \\ &= \frac{18x}{100} \\ &= \frac{9x}{50} \end{aligned}$$

$$P_2 = ₹ 10,800 - x$$

$$R_2 = 20$$

$$T_2 = 1 \text{ year}$$

$$\begin{aligned} \text{S.I.} &= \left[\frac{P \times R \times T}{100} \right] \\ &= \left[\frac{(10,800 - x) \times 20 \times 1}{100} \right] \end{aligned}$$

$$= \frac{20(10,800 - x)}{100}$$
$$= \frac{(10,800 - x)}{5}$$

Total amount = ₹ 2,060

$$\Rightarrow \frac{9x}{50} + \frac{(10,800 - x)}{5} = 2,060$$
$$\Rightarrow \frac{9x}{50} + \frac{10(10,800 - x)}{50} = 2,060$$
$$\Rightarrow \frac{9x}{50} + \frac{(1,08,000 - 10x)}{50} = 2,060$$
$$\Rightarrow \frac{(9x + 1,08,000 - 10x)}{50} = 2,060$$

$$\Rightarrow 1,08,000 - x = 2,060 \times 50$$

$$\Rightarrow 1,08,000 - x = 1,03,000$$

$$\Rightarrow x = 1,08,000 - 1,03,000$$

$$\Rightarrow x = 5,000$$

Other part = ₹ 10,800 - ₹ 5,000

$$= ₹ 5,800$$

Hence, the two parts are ₹ 5,000 at 18% and ₹ 5,800 at 20%.

Question 5

Find the amount and the compound interest on ₹ 16,000 for 3 years at 5% per annum compounded annually.

Answer

Given:

$$P = ₹ 16,000$$

$$R = 5\%$$

$$n = 3 \text{ years}$$

$$\begin{aligned} A &= P \left[1 + \frac{R}{100} \right]^n \\ &= 16,000 \left[1 + \frac{5}{100} \right]^3 \\ &= 16,000 \left[1 + \frac{1}{20} \right]^3 \\ &= 16,000 \left[\frac{20}{20} + \frac{1}{20} \right]^3 \\ &= 16,000 \left[\frac{(20 + 1)}{20} \right]^3 \\ &= 16,000 \left[\frac{21}{20} \right]^3 \\ &= 16,000 \left[\frac{9,261}{8,000} \right] \end{aligned}$$

$$= \left[\frac{14,81,76,000}{8,000} \right]$$

$$= 18,522$$

Also

Compound Interest = Final amount - Original Principal

$$= ₹18,522 - ₹16,000$$

$$= ₹2,522$$

Hence, amount = ₹ 18,522 compound interest = ₹ 2,522.

Question 6

Find the amount and the compound interest on ₹ 20,000 for $1\frac{1}{2}$ years at 10% per annum compounded half-yearly.

Answer

Given:

$$P = ₹ 20,000$$

$$R = 10\%$$

$$n = 1\frac{1}{2} \text{ years}$$

$$= \frac{3}{2} \text{ years}$$

Since, interest is compounded half-yearly,

$$\begin{aligned}A &= P \left[1 + \frac{R}{2 \times 100} \right]^{n \times 2} \\&= 20,000 \left[1 + \frac{10}{200} \right]^{2 \times \frac{3}{2}} \\&= 20,000 \left[1 + \frac{1}{20} \right]^3 \\&= 20,000 \left[\frac{20}{20} + \frac{1}{20} \right]^3 \\&= 20,000 \left[\frac{(20 + 1)}{20} \right]^3 \\&= 20,000 \left[\frac{21}{20} \right]^3 \\&= 20,000 \left[\frac{9,261}{8,000} \right] \\&= \left[\frac{18,52,20,000}{8,000} \right] \\&= 23,152.50\end{aligned}$$

Also

Compound Interest = Final amount - Original Principal

$$= ₹23,152.50 - ₹20,000$$

$$= ₹3,152.50$$

Hence, amount = ₹ 23,152.50 and compound interest = ₹ 3,152.50.

Question 7

Find the amount and the compound interest on ₹ 32,000 for 1 year at 20% per annum compounded half-yearly.

Answer**Given:**

$$P = ₹ 32,000$$

$$R = 20\%$$

$$n = 1 \text{ year}$$

Since, interest is compounded half-yearly,

$$A = P \left[1 + \frac{R}{2 \times 100} \right]^{n \times 2}$$

$$= 32,000 \left[1 + \frac{20}{200} \right]^{2 \times 1}$$

$$= 32,000 \left[1 + \frac{1}{10} \right]^2$$

$$= 32,000 \left[\frac{10}{10} + \frac{1}{10} \right]^2$$

$$= 32,000 \left[\frac{(10 + 1)}{10} \right]^2$$

$$= 32,000 \left[\frac{11}{10} \right]^2$$

$$= 32,000 \left[\frac{121}{100} \right]$$

$$= \left[\frac{38,72,000}{100} \right]$$

$$= 38,720$$

Also

Compound Interest = Final amount - Original Principal

$$= ₹38,720 - ₹32,000$$

$$= ₹6,720$$

Hence, amount = ₹ 38,720 and compound interest = ₹ 6,720.

Question 8

Find the amount and the compound interest on ₹ 4,000 in 2 years, if the rate of interest for first year is 10% and for the second year is 15%.

Answer

Given:

$$P = ₹ 4,000$$

$$T = 2 \text{ years}$$

$$R_1 = 10\%$$

$$R_2 = 15\%$$

$$\text{As we know, } A = P \left[1 + \frac{R_1}{100} \right] \left[1 + \frac{R_2}{100} \right]$$

$$= 4,000 \left[1 + \frac{10}{100} \right] \left[1 + \frac{15}{100} \right]$$

$$= 4,000 \left[1 + \frac{1}{10} \right] \left[1 + \frac{3}{20} \right]$$

$$= 4,000 \left[\frac{10}{10} + \frac{1}{10} \right] \left[\frac{20}{20} + \frac{3}{20} \right]$$

$$= 4,000 \left[\frac{(10 + 1)}{10} \right] \left[\frac{(20 + 3)}{20} \right]$$

$$= 4,000 \left[\frac{11}{10} \right] \left[\frac{23}{20} \right]$$

$$= \left[\frac{10,12,000}{200} \right]$$

$$= ₹5,060$$

$$\text{C.I.} = A - P$$

$$= ₹5,060 - ₹4,000$$

$$= ₹1,060$$

Hence, the amount = ₹ 5,060 and the compound interest = ₹ 1,060.

Question 9

Find the amount and the compound interest on ₹ 10,000 in 3 years, if the rates of interest for the successive years are 10%, 15% and 20% respectively.

Answer**Given:**

$$P = ₹ 10,000$$

$$T = 3 \text{ years}$$

$$R_1 = 10\%$$

$$R_2 = 15\%$$

$$R_3 = 20\%$$

As we know,

$$\begin{aligned} A &= P \left[1 + \frac{R_1}{100} \right] \left[1 + \frac{R_2}{100} \right] \left[1 + \frac{R_3}{100} \right] \\ &= 10,000 \left[1 + \frac{10}{100} \right] \left[1 + \frac{15}{100} \right] \left[1 + \frac{20}{100} \right] \\ &= 10,000 \left[1 + \frac{1}{10} \right] \left[1 + \frac{3}{20} \right] \left[1 + \frac{1}{5} \right] \\ &= 10,000 \left[\frac{10}{10} + \frac{1}{10} \right] \left[\frac{20}{20} + \frac{3}{20} \right] \left[\frac{5}{5} + \frac{1}{5} \right] \\ &= 10,000 \left[\frac{(10+1)}{10} \right] \left[\frac{(20+3)}{20} \right] \left[\frac{(5+1)}{5} \right] \\ &= 10,000 \left[\frac{11}{10} \right] \left[\frac{23}{20} \right] \left[\frac{6}{5} \right] \end{aligned}$$

$$= \left[\frac{1,51,80,000}{1000} \right]$$

$$= 15,180$$

$$\text{C.I.} = A - P$$

$$= ₹15,180 - ₹10,000$$

$$= ₹5,180$$

Hence, the amount = ₹ 15,180 and the compound interest = ₹ 5,180.

Question 10

A sum of money lent at simple interest amounts to ₹ 3,224 in 2 years and ₹ 4,160 in 5 years. Find the sum and the rate of interest.

Answer

Given:

Amount in 2 years = ₹ 3,224

$$\Rightarrow P + \text{S.I. of 2 years} = ₹ 3,224 \dots\dots\dots(1)$$

Amount in 5 years = ₹ 4,160

$$\Rightarrow P + \text{S.I. of 5 years} = ₹ 4,160 \dots\dots\dots(2)$$

Subtracting equation (1) from (2), we get

$$\text{S.I. of 3 years} = ₹ 4,160 - ₹ 3,224 = ₹ 936$$

$$\text{S.I. of 1 year} = ₹ \frac{936}{3} = ₹ 312$$

$$\text{S.I. of 2 years} = ₹312 \times 2 = ₹624$$

From equation (1), we get:

$$P + ₹624 = ₹3,224$$

$$P = ₹3,224 - ₹624$$

$$P = ₹2,600$$

Now when $P = ₹2,600$

$$\text{S.I.} = ₹312$$

$$T = 1 \text{ year}$$

Let the rate be r .

As we know,

$$\text{S.I.} = \left(\frac{P \times R \times T}{100} \right)$$

$$\Rightarrow 312 = \left(\frac{2,600 \times r \times 1}{100} \right)$$

$$\Rightarrow 312 = \frac{2,600r}{100}$$

$$\Rightarrow 312 = 26r$$

$$\Rightarrow r = \frac{312}{26}$$

$$\Rightarrow r = 12$$

Hence, principal = ₹2, 600 and rate = 12

Question 11

At what rate percent per annum compound interest will ₹ 5,000 amount to ₹ 5,832 in 2 years?

Answer

Given:

$$P = ₹ 5,000$$

$$n = 2 \text{ years}$$

$$A = ₹ 5,832$$

Let the rate be r .

As we know,

$$A = P \left[1 + \frac{r}{100} \right]^n$$

$$\Rightarrow 5,832 = 5,000 \left[1 + \frac{r}{100} \right]^2$$

$$\Rightarrow \frac{5,832}{5000} = \left[1 + \frac{r}{100} \right]^2$$

$$\Rightarrow \frac{729}{625} = \left[1 + \frac{r}{100} \right]^2$$

$$\Rightarrow \sqrt{\frac{729}{625}} = \left[1 + \frac{r}{100} \right]$$

$$\begin{aligned}\Rightarrow 1 + \frac{r}{100} &= \frac{27}{25} \\ \Rightarrow \frac{r}{100} &= \frac{27}{25} - 1 \\ \Rightarrow \frac{r}{100} &= \frac{27}{25} - \frac{25}{25} \\ \Rightarrow \frac{r}{100} &= \frac{(27 - 25)}{25} \\ \Rightarrow \frac{r}{100} &= \frac{2}{25} \\ \Rightarrow r &= \frac{2 \times 100}{25} \\ \Rightarrow r &= \frac{200}{25} \\ \Rightarrow r &= 8\end{aligned}$$

Hence, the rate of interest = 8.

Question 12

₹ 16,000 invested at 10% p.a. compounded semi-annually amounts to ₹ 18,522.
Find the time period of investment.

Answer

Given:

$$P = ₹ 16,000$$

$$R = 10\%$$

$$A = ₹ 18,522$$

Let the time be n .

Since the interest is compounded half-yearly,

$$A = P \left[1 + \frac{r}{2 \times 100} \right]^{2 \times n}$$

$$\Rightarrow 18,522 = 16,000 \left[1 + \frac{10}{200} \right]^{2 \times n}$$

$$\Rightarrow \frac{18,522}{16,000} = \left[1 + \frac{10}{200} \right]^{2 \times n}$$

$$\Rightarrow \frac{18,522}{16,000} = \left[1 + \frac{1}{20} \right]^{2 \times n}$$

$$\Rightarrow \frac{18,522}{16,000} = \left[\frac{20}{20} + \frac{1}{20} \right]^{2 \times n}$$

$$\Rightarrow \frac{18,522}{16,000} = \left[\frac{(20 + 1)}{20} \right]^{2 \times n}$$

$$\Rightarrow \frac{18,522}{16,000} = \left[\frac{21}{20} \right]^{2 \times n}$$

$$\Rightarrow \frac{9,261}{8,000} = \left[\frac{21}{20} \right]^{2 \times n}$$

$$\Rightarrow \left[\frac{21}{20} \right]^3 = \left[\frac{21}{20} \right]^{2 \times n}$$

$$\text{Hence, } 2 \times n = 3$$

$$n = \frac{3}{2} \text{ years}$$

$$n = 1\frac{1}{2} \text{ years}$$

Hence, the time period of investment is $1\frac{1}{2}$ years.

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